

Benjamin Schondorf

Path Planning · Scalable Systems · Autonomous Robotics

benschondorf@gmail.com | benjaminschondorf.com | linkedin.com/in/benjamin-schondorf

Skills

Programming Languages: Java, TypeScript, Python, C++, SQL

Frameworks / Tools: Dagger, REST APIs, Java Topology Suite (JTS), Linux, Docker, Redis, ROS2, React

Infrastructure: AWS (Lambda, ECS, DynamoDB, S3, Kinesis, CloudWatch), Infrastructure as Code (CDK)

Experience

Amazon Robotics – North Reading, MA

Feb 2022 – Present

Software Engineer I - II

- Leading redesign of a travel time estimation service, reducing generation time from 5 minutes to 500ms by precomputing on a sparse graph, enabling reachability queries and dynamic obstacle support
- Architected a path data archival service from scratch ingesting 455 million paths/day with sub-500ms ingestion latency, serving 9 consumers including ML pipelines
- Delivered end-to-end integration of 18 robot action types for a cross-team operations service spanning backend, frontend, API, and permissions
- Built path planning regression tests validating changes against all entity-pair combinations and historical production paths in CI/CD
- Mentored 2 co-ops to deliver functioning prototypes and accelerated a new hire's code review cycle from an average of 6 to 2 revisions within one month
- Designed a reusable CDK construct centralizing navigation graph generation, enabling path planning to scale independently of graph generation compute
- Generalized A*-based path planning libraries to support arbitrary entity types via configurable motion profiles and clearance, resolving laden robot path issues in production
- Implemented a traffic lane policy reducing robot conflicts and enabling congestion management in bent corridors, adopted across all production sites

IBM – Austin, TX (Remote)

May 2021 – Aug 2021

Extreme Blue Intern

- Led a 4-person team building a React + Flask application replacing manual SAN storage management, projected to save \$1.45M/year

Ocient – Chicago, IL (Remote)

May 2020 – Aug 2020

Software Engineer Intern

- Added cross-database SQL compatibility in C++ for a petabyte-scale custom database and built static analysis tooling integrated into the CI/CD pipeline

Ford Motor Company – Palo Alto, CA

May 2019 – Aug 2019

Autonomous Vehicle Software Intern

- Built a ROS2 messaging pipeline between autonomous vehicle modules, replacing rate-limited CAN bus communication
- Created a C++ data pipeline interpreting lane-level map data for driving policy algorithms and calculated vehicle localization using lane polynomials

Education

University of Michigan – BS in Computer Science

Dec 2021